HIERARCHICAL INHERITANCE

Java Practical Program

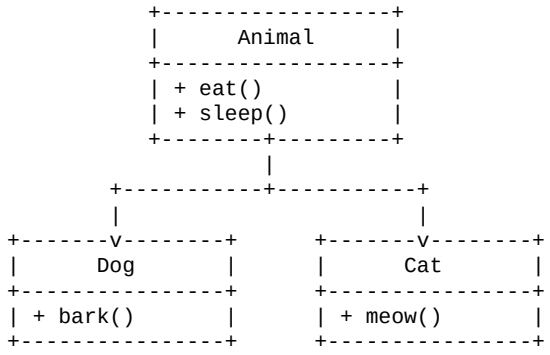
1. Aim

To write a Java program to demonstrate hierarchical inheritance, where multiple child classes inherit properties and methods from a single parent class.

2. Software Requirements

Operating System	Windows / Linux
Programming Language	Java
JDK	JDK 8 or above
IDE	Eclipse / IntelliJ IDEA / NetBeans / VS Code

3. UML Diagram



Dog and Cat both inherit from the parent class Animal. This is hierarchical inheritance.

4. Algorithm

1. Start the program.
2. Create a parent class Animal.
3. Define common methods eat() and sleep() in Animal.
4. Create a child class Dog that extends Animal.
5. Define the bark() method in Dog.
6. Create another child class Cat that extends Animal.
7. Define the meow() method in Cat.
8. Create objects of Dog and Cat.
9. Call inherited and child-class methods.
10. Display the output.
11. Stop the program.

5. Java Program / Code

```
// Hierarchical Inheritance Program

class Animal {
    void eat() {
        System.out.println("Animal eats food");
    }

    void sleep() {
        System.out.println("Animal sleeps");
    }
}

class Dog extends Animal {
    void bark() {
        System.out.println("Dog barks");
    }
}
class Cat extends Animal {
```

```
    void meow() {
        System.out.println("Cat meows");
    }
}

public class HierarchicalInheritance {
    public static void main(String[] args) {

        Dog d = new Dog();
        Cat c = new Cat();

        System.out.println("Dog:");
        d.eat();
        d.sleep();
        d.bark();

        System.out.println();

        System.out.println("Cat:");
        c.eat();
        c.sleep();
        c.meow();
    }
}
```

6. Output

```
Dog:
Animal eats food
Animal sleeps
Dog barks

Cat:
Animal eats food
Animal sleeps
Cat meows
```

7. Output Screenshot

The black-background output above represents the terminal/console screenshot format.

8. Result

Thus, the Java program to demonstrate hierarchical inheritance was successfully executed and the expected output was obtained.